CTMC Models for Customer Journey Prediction

UCLA Stats 148 Capstone — Technical Reference

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May 2026

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1 Problem Setup

Each customer journey is a timestamped sequence of events. The project label is binary:

$$y = \begin{cases} 1 & \text{journey reaches } \texttt{order\_shipped}, \texttt{ed\_id}{=}28, \\ 0 & \text{journey fails by taking no customer action for 60 days.} \end{cases}$$

This is not exactly the same target as the original CTMC experiments. The first CTMC models estimated

$$P(T_{\text{success}} \le 60 \text{ days} \mid X(0) = s),$$

the probability of hitting the success state within a fixed horizon. The classification target is better described as

$$P(T_{\text{success}} < T_{\text{inactivity-failure}} \mid X(0) = s),$$

where the failure time occurs when no customer action happens for 60 days.

2 State-Space Choice

The file `data/Event Definitions.csv` mixes customer actions with company or system events. For the inactivity definition, company/system events should not reset the 60-day clock. Therefore the active CTMC state space uses customer-initiated actions only:

$$\{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 18, 19, 22, 23, 25\}.$$

The success event `order_shipped` (`ed_id`=28) is retained as a terminal outcome state, not as a customer action. The timeout-aware models also add an artificial absorbing failure state `-1`.

This design intentionally excludes states such as approval, decline, pending, catalog mail, downpayment received/cleared, fraud review, account activation, and order fulfillment events from the behavioral CTMC state space.

3 Retained Baseline 1: Global CTMC

3.1 Fitting

For consecutive customer-action transitions $(i, j, \Delta t)$, define

$$N_{ij} = \#\{(i, j, \cdot)\}, \qquad T_i = \sum_{(i,j,\Delta t)} \Delta t.$$

The maximum-likelihood CTMC generator is

$$\hat{q}_{ij} = \frac{N_{ij}}{T_i}, \quad i \ne j, \qquad \hat{q}_{ii} = -\sum_{j \ne i} \hat{q}_{ij}.$$

3.2 Prediction Target

The global baseline still predicts the old target:

$$\hat{p}_{\text{global}}(s) = \left[\exp(Q^* \tau)\right]_{s,28}, \qquad \tau = 60 \cdot 86400.$$

Here Q^* makes state 28 absorbing. This is useful as a baseline but it does not explicitly model inactivity failure.

4 Retained Baseline 2: Clustered CTMC

The clustered CTMC groups journeys by prefix behavior and fits one global CTMC per cluster:

$$Q^{(k)} = \texttt{GlobalCTMC.fit}(\mathcal{D}_k).$$

The implementation supports two clustering modes:

Mode	Switch	Notes
KMeans	default	Original baseline; clusters action proportions and state features.
Spectral	`use_spectral_clustering=True`	Nearest-neighbor spectral fit plus KNN assignment for open jour

The clustered CTMC also predicts the old success-hit target unless paired with one of the timeout-aware models below.

5 New Model 1: Timeout-Absorbing CTMC

5.1 Construction

The timeout transition table modifies the training data:

- Customer-action transitions are kept.
- Transitions into `order_shipped`=28 are kept as success.
- Each incomplete journey receives one artificial terminal transition from its last customer-action state to failure state $f = -1$ after 60 days.

A CTMC generator is then fit on this augmented state space. Both success and failure are absorbing.

5.2 Prediction

For transient states $\mathcal{T} = \mathcal{S} \setminus \{28, -1\}$, the probability of eventually absorbing in success before failure solves

$$Q_{\mathcal{T}\mathcal{T}} u = -Q_{\mathcal{T},28}.$$

The model returns u_s for the current state s. This directly targets

$$P(T_{\text{success}} < T_{\text{failure}} \mid X(0) = s).$$

6 New Model 2: Empirical Semi-Markov Timeout Model

The semi-Markov timeout model removes the exponential waiting-time assumption. For each state i, it stores empirical outgoing samples

$$(j_m, \Delta t_m).$$

The value function is updated by

$$V(i) = \frac{1}{M_i} \sum_{m=1}^{M_i} \begin{cases} 1, & j_m = 28 \text{ and } \Delta t_m \leq 60 \text{ days}, \\ 0, & j_m = -1 \text{ or } \Delta t_m > 60 \text{ days}, \\ V(j_m), & \text{otherwise.} \end{cases}$$

Iteration continues to convergence. This estimates success before timeout using the empirical waiting-time distribution rather than CTMC exponential holding times.

7 Active Model Set

The active CTMC comparison now contains exactly two old CTMC methods and two timeout-aware methods:

Model	Target	Status
`GlobalCTMC`	hit success within 60 days	retained old baseline
`ClusteredCTMC`	hit success within 60 days	retained old baseline
`TimeoutAbsorbingCTMC`	success before inactivity failure	new primary CTMC target
`SemiMarkovTimeoutModel`	success before inactivity failure	new non-exponential target

Older experimental CTMC variants such as neural-rate CTMC, XGBoost-rate CTMC, time-stratified CTMC, Laplace-smoothed CTMC, and isotonic CTMC calibration are not part of the active CTMC comparison.

8 Commands

Run the retained CTMC comparison:

```
python -m src.models.ctmc_pipeline all --no-cache
```

Run only spectral clustered CTMC:

```
python -m src.models.ctmc_pipeline clustered --use-spectral-clustering --no-cache
```